

Ji-hong Min

KAIST, Daejeon, Republic of Korea

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github.com/jihongmin63

Curriculum Vitæ _____ September 1, 2026

Research Interest

I am interested in understanding the structure and behavior of complex computational systems — from large-scale software under formal specification to machine learning models — through mathematical modeling, property-based testing, and mechanized reasoning.

Education

KAIST (Korea Advanced Institute of Science and Technology)

Feb. 2024 – Present

Bachelor of Science in *Mathematical Science and Computer Science*

Individually Designed Minor in *Computational Neuroscience*

- Cumulative Grade Average: 97.3% (4.03 / 4.3)

Korea Science Academy of KAIST

Feb. 2021 – Feb. 2024

Gifted Science High School

Publications & Presentations

1. Sukyoung Ryu, Seokhun Jeong, Jaehyun Lee, and **Ji-hong Min**, “Mechanized Specifications for Real-World Programming Languages,” *The 47th ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, Tutorial, June 2026.

Research Experience

Programming Language Research Group @ KAIST

Dec. 2025 – Present

Undergraduate Researcher

Advisor: Prof. *Sukyoung Ryu*

- Identified the absence of automated validation in SpecTec (a mechanized specification framework), and developed two complementary approaches.
 - + Designed a QuickCheck-style property-based testing framework for specification correctness.
 - + Translated SpecTec specifications into term rewriting systems to enable automated reasoning.
- Surveying specification mechanization toolchains toward a normalized cross-tool comparison.
- Building a property-based falsification loop for Lean proof search in which an LLM writes the test generators and the type checker certifies them.

Computational Cognitive Neuroscience Lab

Jun. 2025 – Dec. 2025

Undergraduate Researcher

Advisor: Prof. *Yul HR Kang*

- Proposed a prospect-theory-based RL model for loss-averse exploration behavior and a general Bayesian inference framework unifying fragmented cognitive task paradigms.

Honors and Awards

National Scholarship for Science and Engineering 2024 – Present
Korean government scholarship for outstanding undergraduates in science and engineering.

KAIST Presidential Fellowship 2024 – Present
KAIST fellowship awarded to selected top students.

Skills

Programming Languages Python, C, OCaml, Rocq, SpecTec
Languages Korean (Native), English (TOEFL iBT 98, CEFR C1)

Service & Leadership

CS101: Introduction to Programming (KAIST) Fall 2026
Teaching Assistant
– Led weekly lab sessions for ~ 40 students; graded assignments and exams.

KAIST Times 2024 – 2025
Official student-run newspaper of KAIST and Daejeon Innopolis.
– Covered campus and external current affairs as *Reporter* and *Section Chief*.